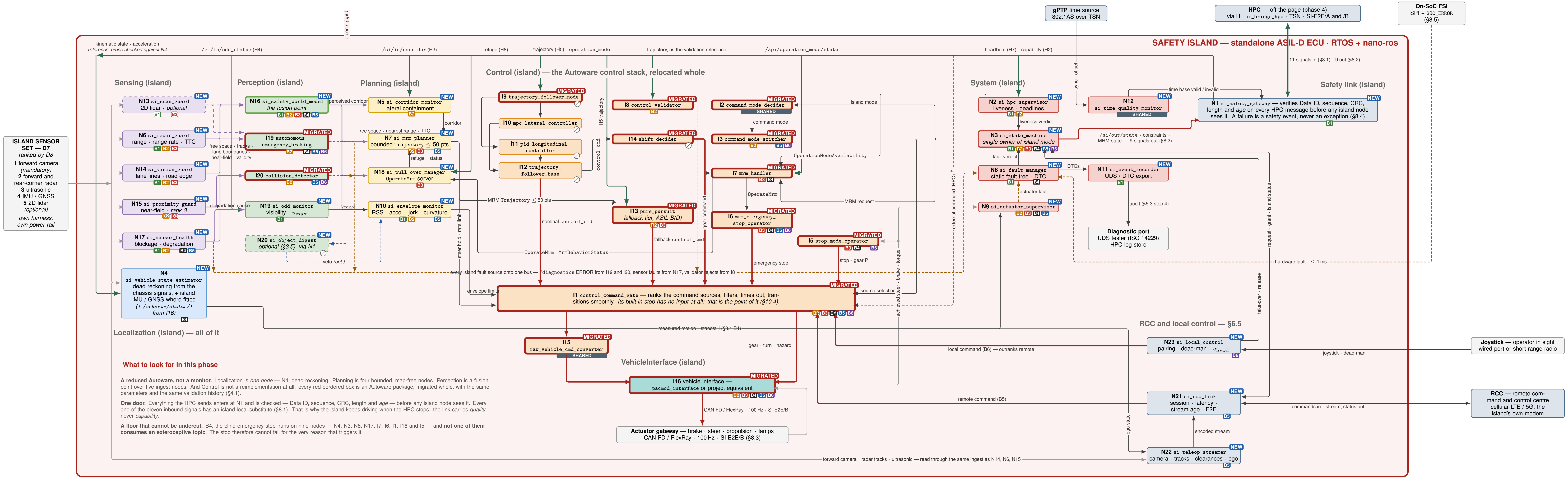
Phase 5 — the Safety Island alone, every node

The last zoom. All 37 island nodes — 18 migrated Autoware **I**-nodes and 19 new **N**-nodes — plus the optional N20, and N21–N23 of §6.5 for B5 and B6, design pending.
Colour is still the Autoware stack, as in `ArchitectureMain.pdf`; the island runs a reduced version of nearly every stack. The HPC is off the page: it reaches this diagram only through N1.



gPTP time source
802.1AS over TSN

HPC — off the page (phase 4)
via H1 `si_bridge_hpc` · TSN · SI-E2E/A and /B

On-SoC FSI
SPI + `SOC_ERROR`
(§8.5)

SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros

ISLAND SENSOR SET — D7
ranked by D8
1 forward camera *(mandatory)*
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar *(optional)*
own harness, own power rail

kinematic state · acceleration
reference, cross-checked against N4

`/si/in/odd_status` (H4) · objects *(opt.)* · `/si/in/corridor` (H3) · refuge (H8) · trajectory (H5) · `operation_mode` · trajectory, as the validation reference · `/api/operation_mode/state` · heartbeat (H7) · capability (H2) · sync · offset · 11 signals in (§8.1) · 9 out (§8.2)

Sensing (island)

- **N13** `si_scan_guard` — 2D lidar · *optional* — NEW — B1 B3
- **N6** `si_radar_guard` — range · range-rate · TTC — NEW — B1 B2 B3
- **N14** `si_vision_guard` — lane lines · road edge — NEW — B1 B2 B3
- **N15** `si_proximity_guard` — near-field · *rank 3* — NEW — B2 B3
- **N17** `si_sensor_health` — blockage · degradation — NEW — B1 B2 B4 B5

Perception (island)

- **N16** `si_safety_world_model` — *the fusion point* — NEW — B1 B2 B3 B4 B5
- **I19** `autonomous_emergency_braking` — MIGRATED — B2 B4 B5 B6
- **I20** `collision_detector` — MIGRATED — B2 B5 B6
- **N19** `si_odd_monitor` — visibility · $v_{\max}$ — NEW — B1
- **N20** `si_object_digest` — *optional (§3.5), via N1* — NEW — ⊘

free space · tracks · lane boundaries · near-field · validity
degradation cause
perceived corridor
$v_{\max}$
veto *(opt.)*

Planning (island)

- **N5** `si_corridor_monitor` — lateral containment — NEW — B2
- **N7** `si_mrm_planner` — bounded `Trajectory` ≤ 50 pts — NEW — B2 B3 B5
- **N18** `si_pull_over_manager` — `OperateMrm` server — NEW — B3
- **N10** `si_envelope_monitor` — RSS · accel · jerk · curvature — NEW — B1 B2 B5

corridor
free space · nearest range · TTC
refuge · status
MRM `Trajectory` ≤ 50 pts
steer hold · rate limit
envelope limits

Control (island) — the Autoware control stack, relocated whole

- **I9** `trajectory_follower_node` — MIGRATED — ⊘
- **I10** `mpc_lateral_controller` — ⊘
- **I11** `pid_longitudinal_controller` — ⊘
- **I12** `trajectory_follower_base` — ⊘
- **I8** `control_validator` — MIGRATED — B2
- **I14** `shift_decider` — MIGRATED — ⊘
- **I13** `pure_pursuit` — *fallback tier, ASIL-B(D)* — MIGRATED — B2 B3
- **I2** `command_mode_decider` — MIGRATED — SHARED
- **I3** `command_mode_switcher` — MIGRATED — B2 B5 B6
- **I7** `mrm_handler` — MIGRATED — B3 B4
- **I6** `mrm_emergency_stop_operator` — MIGRATED — B3 B4 B5 B6
- **I5** `stop_mode_operator` — MIGRATED — B3 B4 B6
- **I1** `control_command_gate` — ranks the command sources, filters, times out, transitions smoothly. *Its built-in stop has no input at all: that is the point of it (§10.4).* — MIGRATED — B2 B3 B4 B5 B6

H5 trajectory
`control_cmd`
nominal `control_cmd`
fallback `control_cmd`
gear command
command mode
island mode
`OperationModeAvailability`
`OperateMrm`
MRM request
emergency stop
stop · gear P
source selection
external command (HPC) †
achieved steer · brake · torque
`OperateMrm` · `MrmBehaviorStatus`
every island fault source onto one bus — `/diagnostics` ERROR from I19 and I20, sensor faults from N17, validator rejects from I8

System (island)

- **N2** `si_hpc_supervisor` — liveness · deadlines — NEW — B1 B2
- **N3** `si_state_machine` — *single owner of island mode* — NEW — B1 B2 B3 B4 B5 B6
- **N8** `si_fault_manager` — static fault tree · DTC — NEW — B1 B4
- **N9** `si_actuator_supervisor` — NEW — B2 B3 B4 B5
- **N12** `si_time_quality_monitor` — NEW — SHARED
- **N11** `si_event_recorder` — UDS / DTC export — NEW — B1

liveness verdict
fault verdict
actuator fault
DTCs
time base valid / invalid
`/si/out/state` · constraints · MRM state — 9 signals out (§8.2)
audit (§5.3 step 4)
hardware fault · ≤ 1 ms

Diagnostic port
UDS tester (ISO 14229)
HPC log store

Safety link (island)

- **N1** `si_safety_gateway` — verifies Data ID, sequence, CRC, length and *age* on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4) — NEW — B1

Localization (island) — all of it

- **N4** `si_vehicle_state_estimator` — dead reckoning from the chassis signals, + island IMU / GNSS where fitted *(+ `/vehicle/status/*` from I16)* — NEW — B4

measured motion · standstill (§3.1 B4)

VehicleInterface (island)

- **I15** `raw_vehicle_cmd_converter` — MIGRATED — SHARED
- **I16** vehicle interface — `pacmod_interface` or project equivalent — MIGRATED — B2 B3 B4 B5 B6

gear · turn · hazard
CAN FD / FlexRay · 100 Hz · SI-E2E/B

Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)

forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15

RCC and local control — §6.5

- **N23** `si_local_control` — pairing · dead-man · v_{local} — NEW — B6
- **N21** `si_rcc_link` — session · latency · stream age · E2E — NEW — B5
- **N22** `si_teleop_streamer` — camera · tracks · clearances · ego — NEW — B5

local command (B6) — outranks remote
remote command (B5)
take-over · release
request · grant · island status
ego state
encoded stream
joystick · dead-man
commands in · stream, status out

Joystick — operator in sight
wired port or short-range radio

RCC — remote command and control centre
cellular LTE / 5G, the island's own modem

What to look for in this phase

A reduced Autoware, not a monitor. Localization is *one node* — N4, dead reckoning. Planning is four bounded, map-free nodes. Perception is a fusion point over five ingest nodes. And Control is not a reimplementation at all: every red-bordered box is an Autoware package, migrated whole, with the same parameters and the same validation history (§4.1).

One door. Everything the HPC sends enters at N1 and is checked — Data ID, sequence, CRC, length and *age* — before any island node sees it. Every one of the eleven inbound signals has an island-local substitute (§8.1). That is why the island keeps driving when the HPC stops: the link carries *quality*, never *capability*.

A floor that cannot be undercut. B4, the blind emergency stop, runs on nine nodes — N4, N3, N8, N17, I7, I6, I1, I16 and I5 — and **not one of them consumes an exteroceptive topic**. The stop therefore cannot fail for the very reason that triggers it.

Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour — B1 B2 B3 B4 B5 B6: **B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED: **SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

B2 B3: **Hollow tag** — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

⊘: **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- → HPC-sourced, verified by N1
- → island sensor data
- → control command into the gate
- ⇢ fault · `/diagnostics`
- → actuator feedback
- → state · mode · decision
- ⇢ optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.